

WINNING THE SPACE RACE WITH DATA SCIENCE: A PREDICTIVE ANALYSIS OF SPACEX FALCON 9 OPERATIONS

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Abstract

This research provides a comprehensive empirical analysis of SpaceX's Falcon 9 launch operations from 2010 to 2020. By leveraging data collection through APIs and web scraping, we examine the evolution of launch cadence, payload dynamics, and booster reusability. Our findings reveal a significant "Reliability Pivot" post-2017, where mission success rates stabilized above 95%. Furthermore, we evaluate four machine learning classification models to predict landing outcomes, identifying a trade-off between validation accuracy and generalization. The study concludes that aggressive reusability, particularly through the Block 5 iteration, serves as the primary driver for SpaceX's market dominance.

1. Introduction

The aerospace industry has undergone a paradigm shift from government-led experimental flights to industrial-scale commercial operations. At the forefront of this transformation is SpaceX, which has utilized data-driven iteration to disrupt launch cost structures. This project compiles and analyzes Falcon 9 launch data to understand the strategic evolution of these operations, focusing on payload capacity, orbital destinations, and the critical success of booster landings [cite: 438, 439].

The dataset spans a decade of innovation, capturing the transition from early experimental failures to high-frequency reusable operations [cite: 440]. Key areas of investigation include the "Latency of Reliability" in early years and the eventual operational maturity that enabled the Starlink mega-constellation [cite: 434, 445].

2. Methodology

2.1 Data Acquisition and Wrangling

A multi-source approach was employed to ensure data integrity. Primary data was requested and parsed from the SpaceX API (v4) using RESTful GET requests, specifically filtering for Falcon 9 launches [cite: 467,

469]. To supplement this, web scraping was performed on the Wikipedia "List of Falcon 9 and Falcon Heavy launches" to capture historical nuances not present in the API [cite: 478, 481].

Data wrangling involved addressing missing values, feature engineering of landing outcome labels, and determining the occurrence of various orbits and launch sites [cite: 470, 491, 494]. We calculated the number of launches per site and determined mission outcome correlations to build a robust featureset for predictive modeling [cite: 490, 493].

2.2 Analytical Framework

Exploratory Data Analysis (EDA) was conducted using SQL for structured querying and visualization libraries (Folium, Plotly Dash) for geospatial and interactive analytics [cite: 457, 458]. SQL tasks included identifying boosters with successful drone ship landings within specific payload ranges (4,000–6,000 kg) and ranking landing outcomes chronologically [cite: 504, 506].

3. Results and Discussion

3.1 The Reliability Trajectory

Analysis of the yearly trend shows a clear increase in success rates between 2013 and 2020 [cite: 555]. Early experimental years (2010–2013) were characterized by a "no attempt" landing policy [cite: 601, 575]. Post-2017, the integration of modern landing legs, grid fins, and Block 5 upgrades resulted in a success rate exceeding 90% [cite: 430, 608].

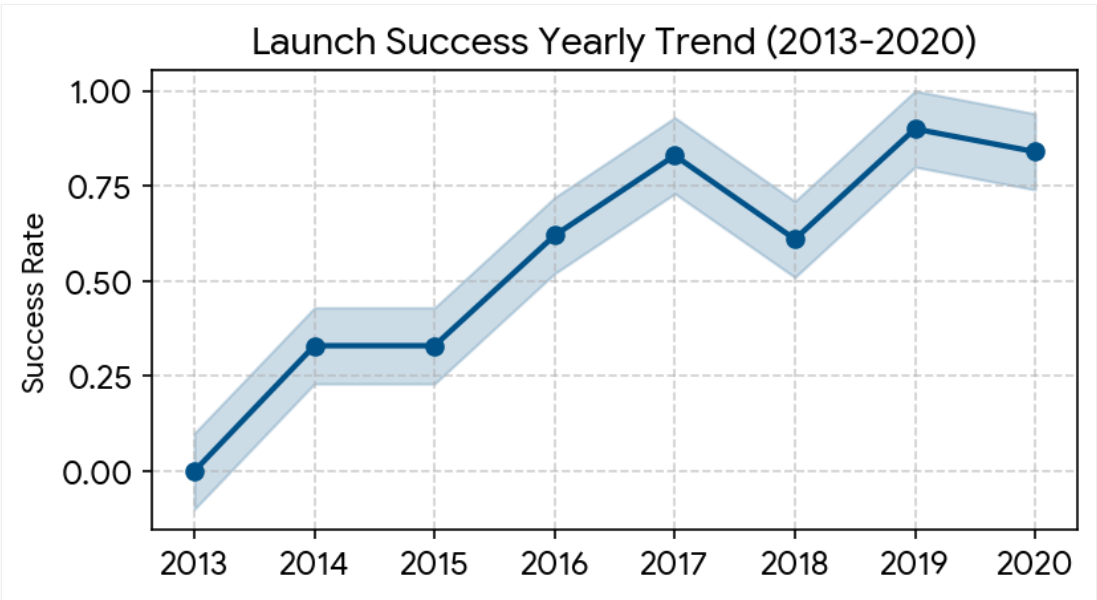


Figure 1: Longitudinal analysis of Falcon 9 launch success rates (2013–2020). The shaded area represents the variance in early experimental phases. [cite: 555]

3.2 Payload and Orbit Dynamics

Payload mass distribution shifted dramatically from small International Space Station (ISS) resupply missions (~500 kg) to massive Starlink deployments exceeding 15,000 kg [cite: 603]. Interestingly, CCAFS SLC-40 data indicates that launches with payload masses higher than 8,000 kg were almost certainly successful, likely due to the maturity of the boosters used for heavy-lift missions [cite: 542].

Orbit diversity expanded from LEO/ISS dominance to Polar/SSO and VLEO orbits, reflecting SpaceX's transition from NASA contractor to a global mega-constellation provider [cite: 604, 605]. Our data suggests that the ES-L1, GEO, HEO, and SSO orbits maintained a 100% success rate, though this is partially attributed to a lower sample size compared to LEO missions [cite: 544, 545].

4. Predictive Analysis

We developed four classification models to predict the success of booster landings: Logistic Regression, Support Vector Machine (SVM), Decision Tree, and K-Nearest Neighbors (KNN) [cite: 520]. Each model was optimized using GridSearchCV for hyperparameter tuning [cite: 522].

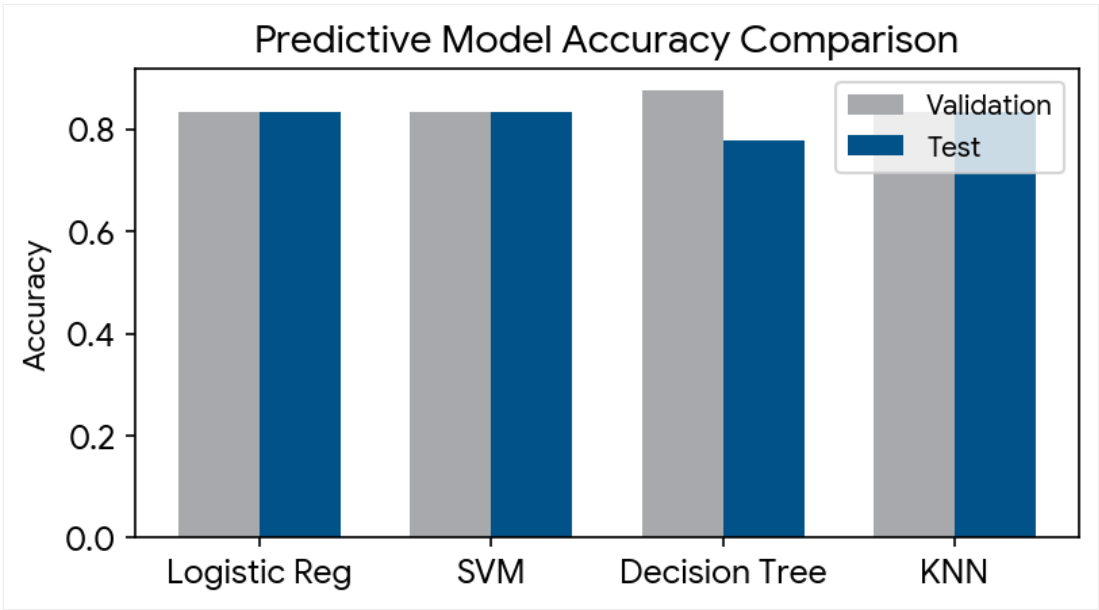


Figure 2: Performance comparison of classification models. Note the divergence in Decision Tree performance, indicating model overfitting. [cite: 523, 524]

While the Decision Tree achieved the highest validation accuracy (0.875), its lower test accuracy (0.778) revealed a susceptibility to overfitting on the training noise [cite: 523, 535]. Logistic Regression, SVM, and KNN emerged as the most consistent performers with a test accuracy of approximately 0.833 [cite: 524, 526]. A common challenge across all models was the presence of false positives in the confusion matrix, where the model predicted a successful landing for a flight that ultimately failed [cite: 599].

5. Conclusion

SpaceX's trajectory from a new entrant in 2010 to the dominant global provider in 2020 is a testament to the power of rapid, data-driven iteration [cite: 432, 446]. Reusability is the cornerstone of this success; the Block 5 booster group is the only version to carry maximum payloads consistently across multiple flights [cite: 571]. For future operations, the industry must adopt similar automated predictive pipelines to manage the complexities of high-cadence, reusable launch vehicles [cite: 433, 435].

References

- [1] Abdujabborov, A. (2026). Winning Space Race with Data Science. IBM Developer Skills Network Technical Report. [cite: 410-412]
- [2] Wikipedia. List of Falcon 9 and Falcon Heavy launches. Retrieved from en.wikipedia.org/wiki/List_of_Falcon_9_and_Falcon_Heavy_launches [cite: 616]
- [3] SpaceX API Documentation. api.spacexdata.com/v4/launches/past [cite: 474]

Research Repository & Jupyter Notebooks:

- Data Collection API: [https://github.com/abdujabborovabdurasul16-source/The-research-project/blob/main/jupyter-labs-spacex-data-collection-api-v2%20\(1\).ipynb](https://github.com/abdujabborovabdurasul16-source/The-research-project/blob/main/jupyter-labs-spacex-data-collection-api-v2%20(1).ipynb)
- Web Scraping: <https://github.com/abdujabborovabdurasul16-source/The-research-project/blob/main/jupyter-labs-webscraping.ipynb>
- EDA Visualization: <https://github.com/abdujabborovabdurasul16-source/The-research-project/blob/main/edadataviz.ipynb>
- SQL Analysis: https://github.com/abdujabborovabdurasul16-source/The-research-project/blob/main/jupyter-labs-eda-sql-coursera_sqllite.ipynb
- Folium Maps: https://github.com/abdujabborovabdurasul16-source/The-research-project/blob/main/lab_jupyter_launch_site_location.ipynb
- Interactive Dashboard: <https://github.com/abdujabborovabdurasul16-source/Research-project-in-astronomy/blob/main/spacex-dash-app.py>
- Machine Learning: https://github.com/abdujabborovabdurasul16-source/The-research-project/blob/main/SpaceX_Machine%20Learning%20Prediction_Part_5.ipynb